



Designing Applications for Gossip Routed Decentralized Networks

Presentation of the Scientific Research Work

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- 3 Implementation & Results
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Gossip Protocol Foundations:

- Peer-to-Peer & Decentralized architecture
- High Fault Tolerance & Resilience
- Propagates messages rapidly [1]
- Ideal for networks assuming no central trust authority (Blockchain, IoT, MANET) [2], [3]

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Identified Drawbacks & Challenges:

- Extreme message redundancy – nodes receive the same message multiple times
- Redundancy consumes critical bandwidth and caps the usable payload size
- Security risk: Large cryptographic shares end up exposed to untrusted intermediate nodes [1]

The Research Gap:

- No prior architecture separates the **cryptographic payload** from the **gossip routing layer**;
- Lack of models treating **every intermediate node** as potentially malicious (Zero-Trust requirement);
- Need to mitigate high **message redundancy** inherent to traditional gossip protocols.

The Research Gap:

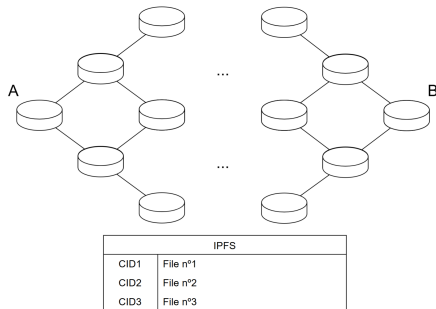
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Proposed Research Aim

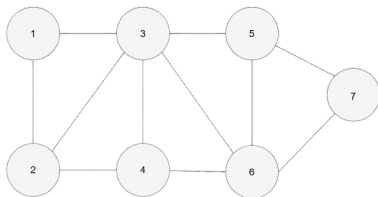
Design and evaluate a decentralized architecture that completely decouples cryptographic payloads from the gossip routing layer using content-addressable storage.

Core Design Pillars:

- Encrypted payload is stored **off-path in IPFS**, outside the gossip routing mesh. [4]
- Only the fixed-size **Content Identifier (CID)** is split via verifiable Shamir's Secret Sharing (VSSS). [5]
- Unique fragments are shared across a simulated network using an adaptive **2-hop gossip protocol**.
- A **probabilistic revocation mechanism** is evaluated to secure dynamic nodes.



2-Hop Adaptive Gossiping (AAG): Forwarding Probability



Probability Dynamics:

- Each node maintains a map of its **2-hop neighborhood**.
- Forwarding probability depends on the availability of redundant routes.
- Sparse nodes** forward with higher probability; **well-connected nodes** scale down. [6], [7]

Mathematical Formulation:

$$P_i = 1 - (1 - \tau_{\text{rel}})^{\frac{1}{|\text{parent}(c_i)|}}$$

$$\tau_{\text{rel}} = \tau_{\text{ARP}}^{1/\delta}$$

Concept: A temporal "kill-switch" utilizing prioritized network propagation to neutralize compromised data.

- **Trigger & Injection:** Upon breach detection, the originator node (Node A) broadcasts a REVOKE command, acting as an "anti-rumor".
- **Priority Routing:** The anti-rumor entirely bypasses Advanced Adaptive Gossiping (AAG) probability computations ($P = 1.0$) and is forwarded with minimal latency.
- **Hop-by-Hop Purge:** As the command arrives, each receiving node instantly deletes locally stored shares and permanently halts further forwarding.

Primary Objective

To mathematically outpace standard fragment propagation and **starve the network** of the k threshold before any downstream node can reconstruct the secret.

Core System Architecture:

- Docker Network Simulation
- Go Routing Logic
- IPFS (Kubo)

Topology	Nodes	n	k	Revocation	Int. Recon.	Secret Reconstructed
Linear	10	2	2	No	No	1
Linear	10	2	2	Yes	No	0
Ring	10	2	2	No	No	1
Ring	10	2	2	Yes	No	1
Ring	20	2	2	No	No	1
Ring	20	2	2	Yes	No	1
Mesh	15	3	2	No	No	7
Mesh	15	3	2	Yes	No	1
Mesh	20	4	3	No	No	2
Mesh	20	4	3	Yes	No	1
Bottleneck	10	3	2	No	Yes	4
Bottleneck	10	3	2	Yes	Yes	2
Bottleneck	15	4	3	No	Yes	2
Bottleneck	15	4	3	Yes	Yes	2

- **Architectural Efficiency:** Offloading the payload to IPFS drastically reduces network overhead without risking data exposure.
- **Optimized Confidentiality:** Combining Dynamic VSSS with AAG routing cuts down message redundancy while keeping the payload secure.
- **Revocation Feasibility:** The priority anti-rumor successfully intercepts secrets, as long as the network has sufficient routing depth.
- **System Limitations:** The design is vulnerable to replication attacks and the newly identified **Multipath Convergence Vulnerability** in bottleneck topologies.

- **Dynamic Bottleneck Mitigation:** Update the AAG protocol to adjust routing in real-time and bypass structural convergence points.
- **Reputation-Based Routing:** Detect, penalize, and route around malicious nodes to stop replication and retention attacks.
- **Metadata Obfuscation:** Integrate techniques like Onion Routing so intermediate nodes cannot correlate fragment data.

- [1] K. Birman, “The promise, and limitations, of gossip protocols,” *ACM SIGOPS Operating Systems Review*, vol. 41, no. 5, pp. 8–13, 2007.
- [2] A. Nadeem and M. P. Howarth, “A survey of manet intrusion detection & prevention approaches for network layer attacks,” *IEEE communications surveys & tutorials*, vol. 15, no. 4, pp. 2027–2045, 2013.
- [3] G. Saldamli, C. Upadhyay, D. Jadhav, R. Shrishrimal, B. Patil, and L. Tawalbeh, “Improved gossip protocol for blockchain applications,” *Cluster Computing*, vol. 25, no. 3, pp. 1915–1926, 2022.
- [4] C. Bieri, “An overview into the interplanetary file system (ipfs): Use cases, advantages, and drawbacks,” *Communication Systems XIV*, vol. 28, 2021.
- [5] C. Georgiou, S. Gilbert, and D. R. Kowalski, “Confidential gossip,” in *2011 31st International Conference on Distributed Computing Systems*, 2011, pp. 603–612. DOI: [10.1109/ICDCS.2011.71](https://doi.org/10.1109/ICDCS.2011.71).
- [6] M. Burmester, T. Van Le, and A. Yasinsac, “Adaptive gossip protocols: Managing security and redundancy in dense ad hoc networks,” *Ad Hoc Networks*, vol. 5, no. 3, pp. 313–323, 2007.
- [7] B. Bako, F. Kargl, E. Schoch, and M. Weber, “Advanced adaptive gossiping using 2-hop neighborhood information,” in *IEEE GLOBECOM 2008-2008 IEEE Global Telecommunications Conference*, IEEE, 2008, pp. 1–6.



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